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# Hyperconvexity and domains of discontinuity for surface groups in $SL_3\mathbb{C}$

BY

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THESIS [PhD candidates may use DISSERTATION here instead]

Submitted as partial fulfillment of the requirements
for the degree of Doctorate of Philosophy in Mathematics
in the Graduate College of the
University of Illinois Chicago, 2026

Chicago, Illinois

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## Accessibility Statement

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# Dedication

For my grandmother Martha, who also saw the joy in geometry.

# Acknowledgements

Emily (integrity, knowledge, high standards)
Sam (love, support, meals, no pressure)
Family (you don't know why I wanted to do this, but that's ok)
Friends (kept me company, kept me happy)
Other math teachers and mentors
Gertrude (tiniest companion)

# Preface

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# List of Abbreviations

- $\mathbb{R}$ The real numbers
- $\mathbb{C}$ The complex numbers

# Summary

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CHAPTER 1

# Introduction

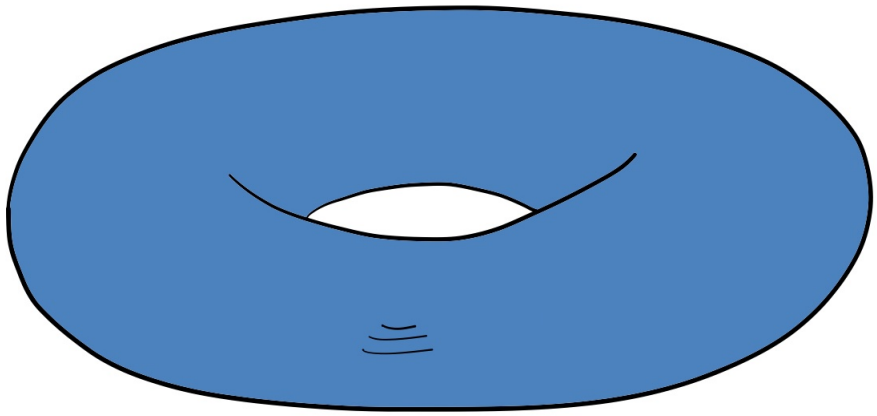

FIGURE 1. This is a torus

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & A' \arrow[r, "f'"] \arrow[d, "a"'] & B' \arrow[r, "g'"] \arrow[d, "b"'] & C' \arrow[r] \arrow[d, "c"'] & 0 \\
0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "a''"'] & B \arrow[r, "g"] \arrow[d, "b''"'] & C \arrow[r] \arrow[d, "c''"'] & 0 \\
0 \arrow[r] & A'' \arrow[r, "f''"] \arrow[d] & B'' \arrow[r, "g''"] \arrow[d] & C'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

FIGURE 2. The Snake Lemma

This is the introduction chapter. We cite some classic works [**1, 2**].

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Sed porttitor urna ac ipsum rhoncus, vel dictum risus commodo. Integer ultrices dui sit amet pulvinar tempor. Phasellus nec tellus eu turpis sodales accumsan. Proin ultr

THEOREM 1.1. *This is a theorem*

We reference Theorem 1.1.

University of Illinois Chicago.

## 1. Motivation

**1.1. Historical context.** A brief overview of how the problem developed.

$$\int_0^1 f(x)dx = 2 \tag{1}$$

How to solve (1)

**1.2. Open questions.** Some questions remain open for future work.
Note: I have not tested the accessibility of this table.

| Monkeys | Lions |
|---|---|
| 100 | 200 |

Table 1. Example Table

CHAPTER 2

# Background

This chapter gives necessary background.

## 1. Group theory

A group is a set $G$ with a binary operation satisfying closure, associativity, identity, and inverses.

THEOREM 2.1. *Every finite subgroup of the multiplicative group of a field is cyclic.*

PROOF. This is a standard result from algebra. $\square$

CHAPTER 3

# Main Result

## 1. Notation

(This maybe belongs in an introduction / background section instead.)

Let $\Gamma$ the fundamental group of a closed hyperbolic surface. Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ Anosov. We use the following notation related to $\rho$ and $\rho'$.

- $\xi, \xi' : \partial\Gamma \to \mathcal{F}^3$ refers to the continuous, transverse, $\rho$- or $\rho'$- equivariant, dynamics-preserving boundary maps.
- $\Lambda, \Lambda' \subset \mathcal{F}^3$ refers to the limit curves, which are the images of $\xi, \xi'$.
- $Th(\Lambda), Th(\Lambda')$ refers to the thickened limit curve.
- $Th(x), Th(x')$ refer to the thickening of specific flags $x \in \Lambda, x' \in \Lambda'$.

For any flag $x \in \mathcal{F}^3$, consider the Schubert varieties associated with $x$. We use the notation $x = (l, H)$ where $x_1 \in \mathbb{CP}^2$ and $x_2 \in (\mathbb{CP}^2)^*$ such that $x_1 \subset x_2$.

- $\{x\} = \{(l, H)\}$,
- $\alpha_1(x) = \{(l, H_l) : l \subset H_l\}$,
- $\alpha_2(x) = \{(l_H, H) : l_H \subset H\}$,
- $\alpha_{12}(x) = \{(k, H_l) : l, k \subset H_l\}$,
- $\alpha_{21}(x) = \{(l_H, G) : l_H \subset H, G\}$, and
- $\alpha_{121}(x) = \alpha_{212}(x) = \mathcal{F}^3$.

We call two flags $x, y$ transverse if they are in general position, meaning that for any proper Schubert variety $\alpha$, then $x \notin \alpha(y)$ and $y \notin \alpha(x)$.

For any pair of transverse flags $x, y \in \mathcal{F}^3$, consider the irreducible 0-dimensional Richardson varieties $R^0_{x,y}$. There are exactly six, $\{x, x_y, x^y, y_x, y^x, y\}$ where:

- $x = \{x\} \cap \alpha_{121}(y)$,
- $x_y = \alpha_1(x) \cap \alpha_{12}(y)$,
- $x^y = \alpha_2(x) \cap \alpha_{21}(y)$,
- $y_x = \alpha_{12}(x) \cap \alpha_1(y)$,
- $y^x = \alpha_{21}(x) \cap \alpha_2(y)$, and
- $y = \alpha_{121}(x) \cap \{y\}$.

Example 3.1. *Suppose $x$ and $y$ are the attracting and repelling flags of $\rho(g) = diag(\lambda, \mu, \nu)$ where $|\lambda| > |\mu| > |\nu|$. Then $x = (e_1, e_1 + e_2)$, $x_y = (e_1, e_1 + e_3)$, $x^y = (e_2, e_1 + e_2)$, $y_x = (e_3, e_1 + e_3)$, $y^x = (e_2, e_2 + e_3)$, and $y = (e_3, e_2 + e_3)$.*

## 2. Statement of Theorem 3.1 and related results

We call $\rho, \rho'$ *topologically conjugate* if there exists a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that for all $g \in \Gamma$ and all $x \in \mathcal{F}^3$, then $F(\rho(g) \cdot x) = \rho'(g) \cdot F(x)$. We call $F$ a *topological conjugacy.*

Theorem 3.1. *Hyperconvexity is closed under topological conjugacy. In other words, if $\rho$ and $\rho'$ are topologically conjugate, then $\rho$ is hyperconvex if and only if $\rho'$ is hyperconvex.*

The proof of this theorem uses two lemmas, in which we assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between $\rho$ and $\rho'$. Clearly $F(\Lambda) = \Lambda'$. This mapping is pointwise, i.e. for all $\alpha \in \partial\Gamma$, then $F(\xi(\alpha)) = \xi'(\alpha)$. So for $x \in \Lambda$, we can use the notation $x' := F(x) \in \Lambda'$.

The relative position of 0-dimensional Richardson varieties of $\Lambda$ determines whether $\rho$ is hyperconvex.

LEMMA 3.1.1. *The representation $\rho$ is hyperconvex if and only if for all $x, y, z \in \Lambda^{(3)}$, then $R^0_{x,y} \cap R^0_{x,z} = \{x\}$.*

Topological conjugacies preserve the 0-dimensional Richardson varieties, almost point-by-point.

LEMMA 3.1.2. *For all x,y $\in \Lambda^{(2)}$,Fsends$R^0_{x,y}$ to $R^0_{x',y'}$.*

- *If $F$* matches *Richardson varieties, then $F(x_y) = x'_{y'}$, $F(x^y) = x'^{y'}$.*
- *If $F$* switches *Richardson varieties, then $F(x_y) = x'^{y'}$, $F(x^y) = x'_{y'}$.*

As a consequence of the argument in Lemma 3.1.2, we also have the following result:

COROLLARY 3.1.1. *For all $x \in \Lambda$, then $F(Th(x)) = Th(x')$.*

## 3. Proof of Theorem 3.1.

We first prove our main theorem using our supporting lemmas.

PROOF. (Theorem 3.1) Suppose $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between $\rho$ and $\rho'$.

We begin with the forward direction. Assume that $\rho$ is hyperconvex. Let $x, y, z \in \Lambda^{(3)}$. By Lemma 3.1.1, then $x_y \neq x_z$ and $x^y \neq x^z$. By Lemma 3.1.2, then $\{F(x_y), F(x^y), F(x_z), F(x^z)\} = \{x'_{y'}, x'^{y'}, x'_{z'}, x'^{z'}\}$ are distinct. Since $x, y, z$ were general, then this holds for all triples in $\Lambda$ and $\rho'$ is hyperconvex.

The other direction proceeds similarly. Assume that $\rho$ is *not* hyperconvex. By Lemma 3.1.1, there exist $x, y, z \in \Lambda^{(3)}$ such that $x_y = x_z$ or $x^y = x^z$. By Lemma 3.1.2, then $F$ either matches or switches Richardson varieties. So one of the following is true: $x'_{y'} = x'_{z'}$ or $x'^{y'} = x'^{z'}$. Therefore $\rho'$ is *not* hyperconvex. □

We also prove our lemma linking Richardson varieties to hyperconvexity.

PROOF. (Lemma 3.1.1) Suppose $\rho$ is Anosov. Let $x = (x_1, x_2), y = (y_1, y_2), z = (z_1, z_2) \in \Lambda^{(3)}$. Since $x, y, z$ are pairwise transverse, then $y_x, y^x, z_x, z^x \notin R^0_{x,y} \cap R^0_{x,z}$. It suffices to check the pairs $x_y, x_z \in \alpha_1(x)$ and $x^y, x^z \in \alpha_2(x)$.

By definition, $x_y = (x_1, x_1 + y_1)$ and $x_z = (x_1, x_1 + z_1)$, which are distinct if and only if $x_1 + y_1 \neq x_1 + z_1$, or equivalently $z_1 \not\subset x_1 + y_1$.

Similarly, $x^y = (x_2 \cap y_2, x_2)$ and $x^z = (x_2 \cap z_2, x_2)$, which are distinct if and only if $x_2 \cap y_2 \neq x_2 \cap z_2$, or equivalently $x_2 \cap y_2 \not\subset z_2$.

So $R^0_{x,y} \cap R^0_{x,z} = \{x\}$ for all $x, y, z \in \Lambda^{(3)}$ is equivalent to $\rho$ hyperconvex. □

The remainder of this chapter is devoted to the proof of Lemma 3.1.2.

## 4. Action of $\rho(g)$ on $F^3$

We describe the dynamics of $\rho(g) \cdot \mathcal{F}^3$, where $g \in \Gamma$ infinite order. Since $\rho$ is Anosov, then we know $\rho(g)$ is strongly loxodromic, i.e. its eigenvalues have distinct magnitude. That means it has an attracting and repelling fixed boundary points $g^+, g^- \in \partial\Gamma$. Let $x = \xi(g^+)$ and $y = \xi(g^-)$.

Under $\rho(g)$, most of $\mathcal{F}^3$ is attracted to $x$ and repelled by $y$. We define the *uphill of* $x$ as the subset attracted to $x$ under $\rho(g)$:

$$U_{\rho(g)}(x) := \{f \in F^3 : \rho(g^n) \cdot f \to x\}. \tag{2}$$

Similarly, we define the *downhill of* $y$ as the subset repelled by $y$ under $\rho(g)$:

$$D_{\rho(g)}(y) := \{f \in \mathcal{F}^3 : \rho(g^{-n}) \cdot f \to y\}. \tag{3}$$

Finally, we define the *exceptional set* as the points not attracted to $x$ or not repelled by $y$ under $\rho(g)$:

$$E_{\rho(g)} = \mathcal{F}^3 \setminus (U_{\rho(g)}(x) \cap D_{\rho(g)}(y)). \tag{4}$$

The set $E_{\rho(g)}$ is nonempty, since it contains $x$ and $y$, and closed, since it is complement of an open set. It also contains the other fixed points of $\rho(g) \cdot \mathcal{F}^3$, which are precisely $R^0_{x,y}$. (This is straightforward to verify in Example 3.1, which is generic because up to conjugation $\rho(g) = diag(\lambda, \mu, \nu)$.)

We can describe subsets of $E_{\rho(g)}$ explicitly by their attracting and repelling fixed point. These subsets are the irreducible 1-dimensional Richardson varieties $R^1_{x,y}$:

- $span(x, x_y) = \alpha_1(x) = \alpha_1(x) \cap \alpha_{121}(y) = \overline{U(x) \cap D(x_y)}$
- $span(x, x^y) = \alpha_2(x) = \alpha_2(x) \cap \alpha_{121}(y) = \overline{U(x) \cap D(x^y)}$
- $span(x_y, y_x) = \alpha_{12}(x) \cap \alpha_{12}(y) = \overline{U(x_y) \cap D(y_x)}$
- $span(x^y, y^x) = \alpha_{21}(x) \cap \alpha_{21}(y) = \overline{U(x^y) \cap D(y^x)}$
- $span(y_x, y) = \alpha_1(y) = \alpha_{121}(x) \cap \alpha_1(y) = \overline{U(y_x) \cap D(y)}$
- $span(y^x, y) = \alpha_2(y) = \alpha_{121}(x) \cap \alpha_2(y) = \overline{U(y^x) \cap D(y)}$

These varieties are not disjoint, e.g. $\{x\} = span(x, x_y) \cap span(x, x^y)$, but the points of intersection are exactly the 0-dimensional Richardson varieties. So $E_{\rho(g)}$ is homeomorphic to the union of six copies of $\mathbb{CP}^1$, each identified with two neighbors in a cyclic order $(x, x_y, y_x, y, y^x, x^y)$.

For the following results, let $F : \mathcal{F}^3 \to \mathcal{F}^3$ be a topological conjugacy between $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$. We use the notation $x' = F(x), y' = F(y)$.

Proposition 3.1. *Let $g \in \Gamma$, and suppose $x = \xi(g^+), y = \xi(g^-)$. Then $F(x_y) \in \{x'_{y'}, x'^{y'}\}$.*

Proof. Let $S : S^1 \to E_{\rho(g)}$ continuous and injective such that it passes through $(x, x_y, y_x, y, y^x, x^y)$ in cyclic order. Without loss of generality, we may assume $S(0) = x$, $S(\pi/3) = x_y$, etc. Since $F$ is a topological conjugacy, then it must preserve the exceptional set, i.e. $F(E_{\rho(g)}) = E_{\rho'(g)}$.

Then $F \cdot S : S^1 \to E_{\rho'(g)}$ is also injective. We also know $F \cdot S(0) = x'$ and $F \cdot S(\pi) = y'$. So by injectivity and the cyclic order of $R^0_{x',y'}$ in $E_{\rho'(g)}$, then either $F \cdot S(\pi/3) = x'_{y'}$ or $F \cdot S(\pi/3) = x'^{y'}$. The remaining points $F \cdot S(n\pi/3)$ where $n \in \{2, 4, 5\}$ are determined by this choice as well. □

## 5. $F$ preserves Richardson varieties of $\Lambda$

In this section, we still assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between $\rho, \rho'$. We extend our argument from fixed points of infinite order words in $\Gamma$ to generic Richardson varieties in $\Lambda$.

First, we extend Proposition 3.1 to Richardson varieties of the form $x_z$ where $x$ is the attracting fixed point of an infinite order element $g \in \Gamma$ and $z$ is a generic point on $\Lambda$.

Proposition 3.2. *Let $g \in \Gamma$ infinite order, and define $x := \xi(g^+)$. Let $z \in \Lambda \setminus \{x\}$. Then $F(x_z) \in \{x'_{z'}, x'^{z'}\}$.*

We use the following result from group theory:

REMARK 3.1. *Let $\Gamma$ hyperbolic with boundary $\partial\Gamma \equiv S^1$, and let $x, z \in \partial\Gamma$. Then there exists a quasigeodesic sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that for all $y \in \partial\Gamma$:*

$$lim_{i\to\infty} g_i \cdot y = \begin{cases} x : y = x \\ z : y \neq x \end{cases} . \tag{5, 6}$$

PROOF. (Proposition 3.2) If $z = \xi(g^-)$ then we are done, so assume not. Set $y = \xi(g^-)$.

Let $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ as in Rem. 3.1, such that $\lim_{i=1} \rho(g_i) \cdot y = z$ and $lim_{i=1}\rho(g_i) \cdot x = x$. Let $x_i := \rho(g_i) \cdot x$ and $y_i := \rho(g_i) \cdot y$, and let $x'_i := F(x_i), y'_i := F(y_i)$.

By Prop. **??** then for all $i \in \mathbb{N}$ we have $F(x_{iy_i}) \in \{x'_{iy'_i}, x'^{y'_i}_i\}$. Clearly $\lim_{i\to\infty} x_{iy_i} = x_z$, $\lim_{i\to\infty} x'_i = x'$ and $\lim_{i\to\infty} y'_i = z'$. By continuity of $F$ and compactness of $\mathcal{F}^3$, then

$$\begin{aligned} F(x_z) &= F(\lim_{i\to\infty} x_{iy_i}) && (7)\\ &= \lim_{i\to\infty} F(x_{iy_i}) && (8)\\ &\in \{\lim_{i\to\infty} x'_{iy'_i}, \lim_{i\to\infty} x'^{y'_i}_i\} && (9)\\ &= \{x'_{z'}, x'^{z'}\}. && (10)\end{aligned}$$

□

Next, we extend Proposition 3.2 to Richardson varieties of the form $w_z$ where both $w$ and $z$ are generic points of $\Lambda$.

PROPOSITION 3.3. *For all $w, z \in \partial\Gamma^{(2)}$, then $F(w_z) \in \{w'_{z'}, w'^{z'}\}$.*

PROOF. Let $w, z \in \Lambda$. If $w$ is the attracting fixed point of an infinite order element, we are done by Prop 3.2, so assume not.

Let $\{x_i\}_{i=1} \in \Lambda^{\mathbb{N}}$ such that each $x_i = \xi(g_i)$ the attracting fixed point of an infinite order word, and $\lim_{i\to\infty} x_i = w$. This sequence exists because attracting fixed points of $\rho(g)$ (where $g$ infinite order) are dense in $\Lambda$. Define $x'_i = F(x_i)$. By continuity of $F$, we have

$$\begin{aligned} F(w) &= F(\lim_{i\to\infty} x_i) && (11)\\ &= \lim_{i\to\infty} F(x_i) && (12)\\ &= \lim_{i\to\infty} (x'_i) && (13)\\ &= w'. && (14)\end{aligned}$$

Then by Prop 3.2 and by continuity of $F$, we have

$$\begin{aligned} F(w_z) &= F(\lim_{i\to\infty} x_{iz}) && (15)\\ &= \lim_{i\to\infty} F(x_{iz}) && (16)\\ &\in \{\lim_{i\to\infty} (x'_{iz'}), \lim_{i\to\infty} (x'^{z'}_i)\} && (17)\\ &= \{w'_{z'}, w'^{z'}\}. && (18)\end{aligned}$$

□

## 6. $F$ matches or switches Richardson varieties

In this section, we highlight how $F$ preserves the balanced ideal thickening centered at each point $x \in \Lambda$. Observe that $Th(x) = \alpha_1(x) \cup \alpha_2(x) = span(x, x_y) \cup span(x, x^y)$ for some $y \in \Lambda \setminus \{x\}$.

Proof. (Corollary 3.1.1) By Prop. 3.1, $\{F(x_y), F(x), F(x^y)\} = \{x'_{y'}, x, x'^{y'}\}$. Observe that $F$ is a homeomorphism that preserves dynamics in that $F(U_{\rho(g)}) = U_{\rho'(g)}$ and $F(D_{\rho(g)}) = D_{\rho'(g)}$. So then

$$F(Th(x)) = F(span(x, x_y) \cup span(x, x^y)) \tag{19}$$
$$= span(x', x'_{y'}) \cup span(x', x'^{y'}) \tag{20}$$
$$= Th(x'). \tag{21}$$

□

We also show that $F$ either *matches* or *switches* Richardson varieties, meaning that it either sends preserves $\alpha_1$ for each $x \in \Lambda$, or swaps it with $\alpha_2$.

Corollary 3.1.2. *If for some $w \in \Lambda$ we have $F(\alpha_1(w)) = \alpha_1(w')$, then for all $z \in \Lambda$ we have $F(\alpha_1(z)) = \alpha_1(z')$. That is, $F$ either* matches *or* switches *Richardson varieties.*

Proof. Suppose for some $w \in \Lambda$ we have $F(\alpha_1(w)) = \alpha_1(w')$. Let $z \in \Lambda$ $\{w\}$. Since $F$ is a homeomorphism, by Cor. 3.1.1 we have $F(\alpha_1(z)) \in \{\alpha_1(z'), \alpha_2(z')\}$.

For contradiction, suppose that $F(\alpha_1(z)) = \alpha_2(z')$. Then for all $g \in \Gamma$, by the definition of a topological conjugacy, we have $F(\alpha_1(\rho(g) \cdot z)) = \alpha_2(\rho'(g) \cdot z')$.

Since $\rho(g) \cdot z$ is dense in $\Lambda$, then there exists a sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that $\lim_{i\to\infty} \rho(g_i) \cdot z = w$. Define $z_i = \rho(g_i) \cdot z$.

Then

$$\alpha_1(w') = F(\alpha_1(w)) \tag{22}$$
$$= F(\alpha_1(\lim_{i\to\infty} z_i)) \tag{23}$$
$$= \lim_{i\to\infty} F(\alpha_1(z_i)) \tag{24}$$
$$= \lim_{i\to\infty} \alpha_2(w) \tag{25}$$
$$= \alpha_2(w). \tag{26}$$

This is a contradiction. □

Clearly $x_y \in \alpha_1(x)$, so if $F$ matches Richardson varieties then $F(x_y) = x'_{y'}$, and if $F$ switches Richardson varities then $F(x_y) = x'^{y'}$.

So for all $x, y \in \Lambda$, a topological conjugacy $F$ sends $R^0_{x,y}$ to $R^0_{x',y'}$, and either matches or switches the irreducible 0- and 1-dimensional Richardson varieties. This completes the proof of Lemma 3.1.2.

## 7. Comments

Remark 3.2. *For $x, y \in (\mathcal{F}^3)^{(2)}$, $F$ will not necessarily preserve the 0-dimensional Richardson varieties.*

Remark 3.3. *Theorem 3.1 applies to virtual hyperbolic surface groups as well, e.g. $\Gamma = \pi_1(\mathcal{O}^2)$ where $\mathcal{O}^2$ is a hyperbolic 2-orbifold.*

CHAPTER 4

# Another result

## 1. Notation

## 2. Statement of Theorem 4.1 and related results

THEOREM 4.1. *Suppose $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ in the same Anosov component. Suppose furthermore that $\rho$ and $\rho'$ are irreducible. Then $\rho, \rho'$ are topologically conjugate.*

Combined with Theorem 3.1, we get the following corollaries:

COROLLARY 4.1.1. *All quasi-Hitchin representations are hyperconvex. All quasi-Barbot representations are* not *hyperconvex.*

Furthermore:

COROLLARY 4.1.2. *The quasi-Hitchin and quasi-Barbot Anosov components are distinct.*

But what about the reducible representations? Well, we must be a bit more careful here. Consider the reducible Anosov representations $\mathcal{A}_{red}$. The connected components are each homeomorphic to Quasi-Fuchsian space.

THEOREM 4.2. *There are exactly three topological conjugacy classes of $\mathcal{A}_{red}(\Gamma)$:*

- $[\rho_{ss}]$
- $[\rho_{line}] = [\rho_{plane}]$
- $[\rho_{dense}]$

## 3. Proof of Theorem 4.1

We use the following result of Florian Stecker:

THEOREM 4.3 (TODO REFERENCE). *For all $\rho : \Gamma \to SL_3\mathbb{C}$ Anosov, the cocompact domain of discontinuity $\Omega_\rho$ is simply connected.*

We use the following result of Dumas and Sanders:

THEOREM 4.4 (TODO REFERENCE). *For all $\rho, \rho'$ in the same Anosov component, then*

$$M_\rho \equiv_{diffeo} M_{\rho'}. \tag{27}$$

PROOF. (Theorem 4.1) Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ Anosov and in the same component. We want to show the existence of $F : \mathcal{F}^3 \to \mathcal{F}^3$ a topological connjugacy of $\rho, \rho'$.

Then by Thm. 4.4, there exists a diffeomorphism $D : M_\rho \to M_{\rho'}$. This lifts to a diffeomorphism of the universal cover $\tilde{D} : \tilde{M}_\rho \to \tilde{M}_{\rho'}$. Then by Thm. 4.3, the domains of discontinuity $\Omega_\rho$, $\Omega_{\rho'}$ are simply connected, so $\tilde{M}_\rho \equiv \Omega_\rho$ and $\tilde{M}_{\rho'} \equiv \Omega_{\rho'}$. So we can consider $\tilde{D} : \Omega_\rho \to \Omega_{\rho'}$.

Clearly $\tilde{D}$ is a topological conjugacy of $\Omega_\rho$ and $\Omega_{\rho'}$, since it is a diffeomorphism and for all $f \in \Omega_\rho$ and $g \in \Gamma$ we have $\tilde{D}(\rho(g) \cdot f) = \rho'(g) \cdot \tilde{D}(f)$.

Next, we want to extend $\tilde{D}$ to a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that $F|_{\Omega_\rho} \equiv \tilde{D}$. Fortunately, $\Omega_\rho$ is an open, connected dense subset of $\mathcal{F}^3$, so $\tilde{D}$ has a unique continuous extension to $\mathcal{F}^3$ using sequences in $\Omega_\rho$ which limit to points in $Th(\Gamma)$.

We just need to check that $F$ is a topological conjugacy. □

REMARK 4.1. *The topological conjugacy $F$ may not be unique, however it is uniquely determined by the choice of diffeomorphism $D$ and its lift $\tilde{D}$.*

## 4. Proof of Theorem 4.2

CHAPTER 5

# Vitae

For UIC this can be a full CV or a short version (e.g. half a page) with previoius degrees, work experience and publications.

APPENDIX A

# Technical Lemmas

Here we collect some supporting lemmas.

# Bibliography


[1] R. Hartshorne, *Algebraic Geometry*, Springer-Verlag, New York, 1977.

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[3] J. Draisma, E. Horobet, G. Ottaviani, B. Sturmfels, and R. R. Thomas, "The Euclidean distance degree of an algebraic variety," *arXiv:1309.0049* (2013). Available at: https://arxiv.org/abs/1309.0049